

Experiment 1 — Installation of Hypervisors and initiation of VMs with image file.

Install and configure an open-source hypervisor and create/instantiate a Virtual Machine (VM) using an existing image file. Demonstrate the successful installation and execution of the VM.

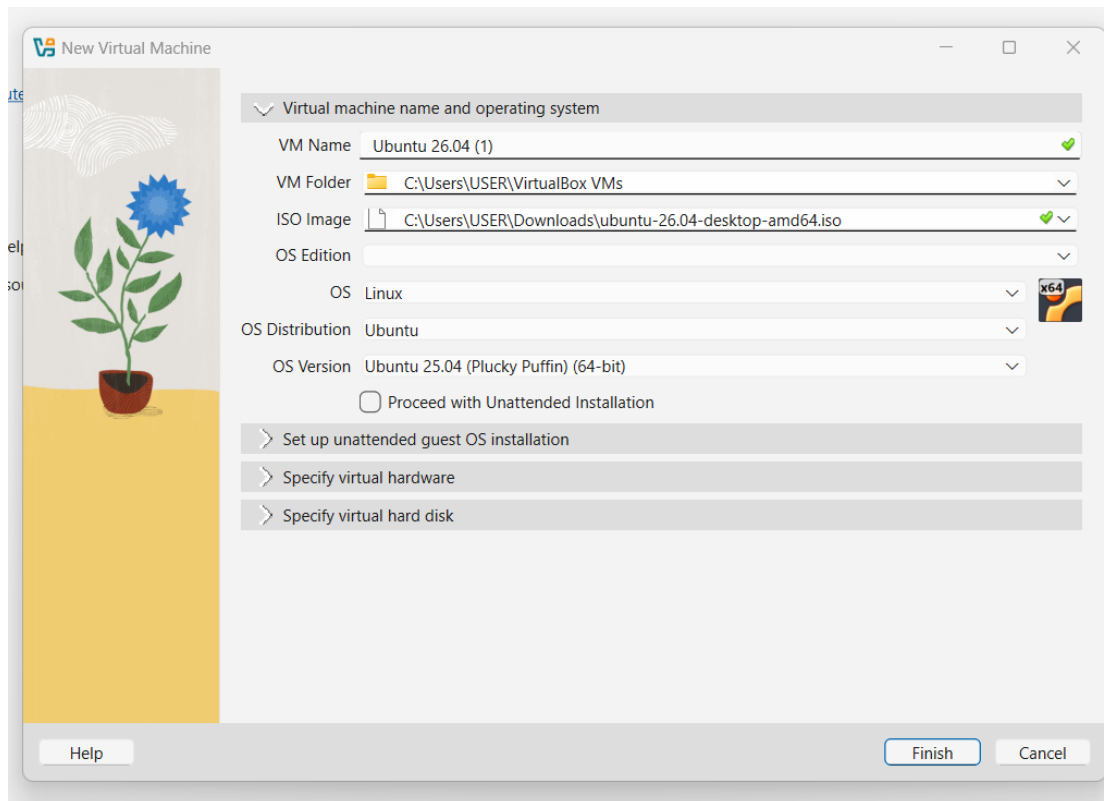
Tasks

- Install an open-source hypervisor such as Oracle VirtualBox
- Obtain a suitable VM image/ISO file.
- Create a new VM using the hypervisor.
- Configure the VM's CPU, RAM, storage, and network.
- Attach the image/ISO file to the VM.
- Start and boot the VM.
- Complete the guest OS installation, if required.
- Verify that the VM is running successfully.
- Installing the required build tools and Linux kernel headers using the below commands:

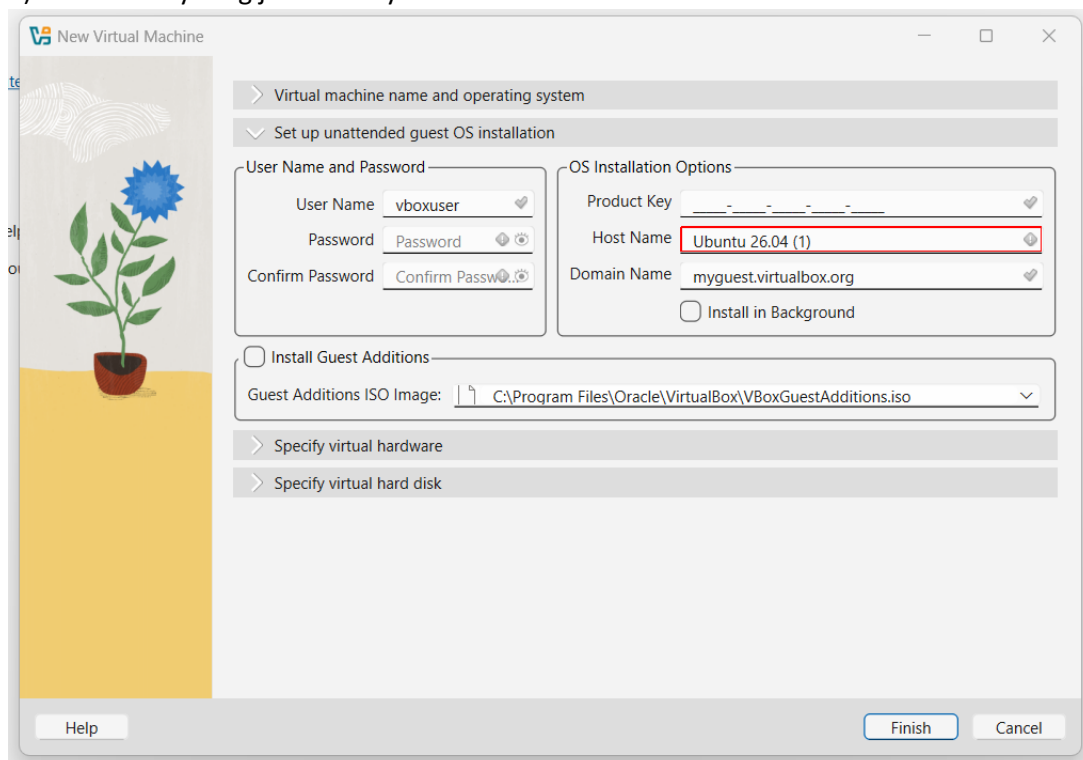
```
sudo apt update
```

```
sudo apt install -y build-essential linux-headers-$(uname -r)
```
- Record the configuration and capture relevant screenshots.

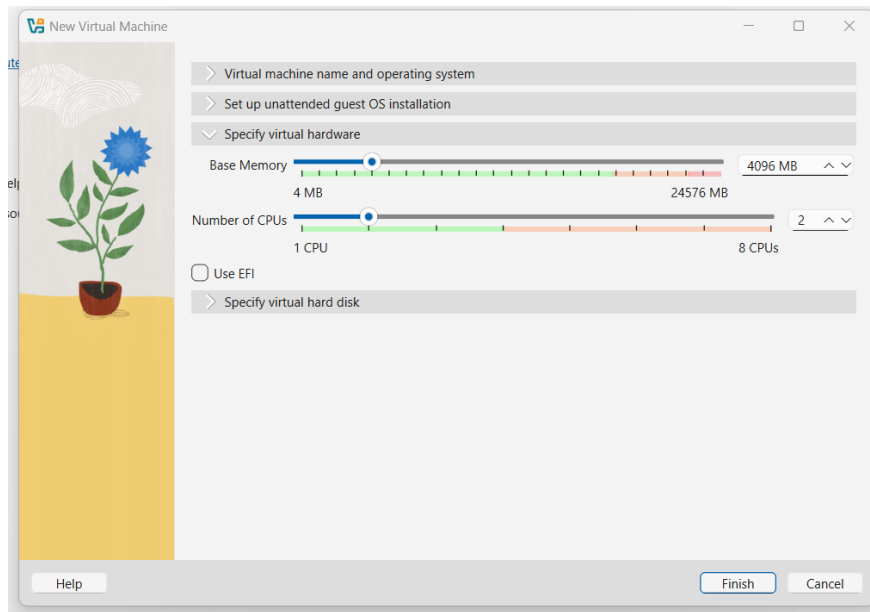
- 1) Now I create VM-1.
- 2) Give VM Name
- 3) Give the ISO file



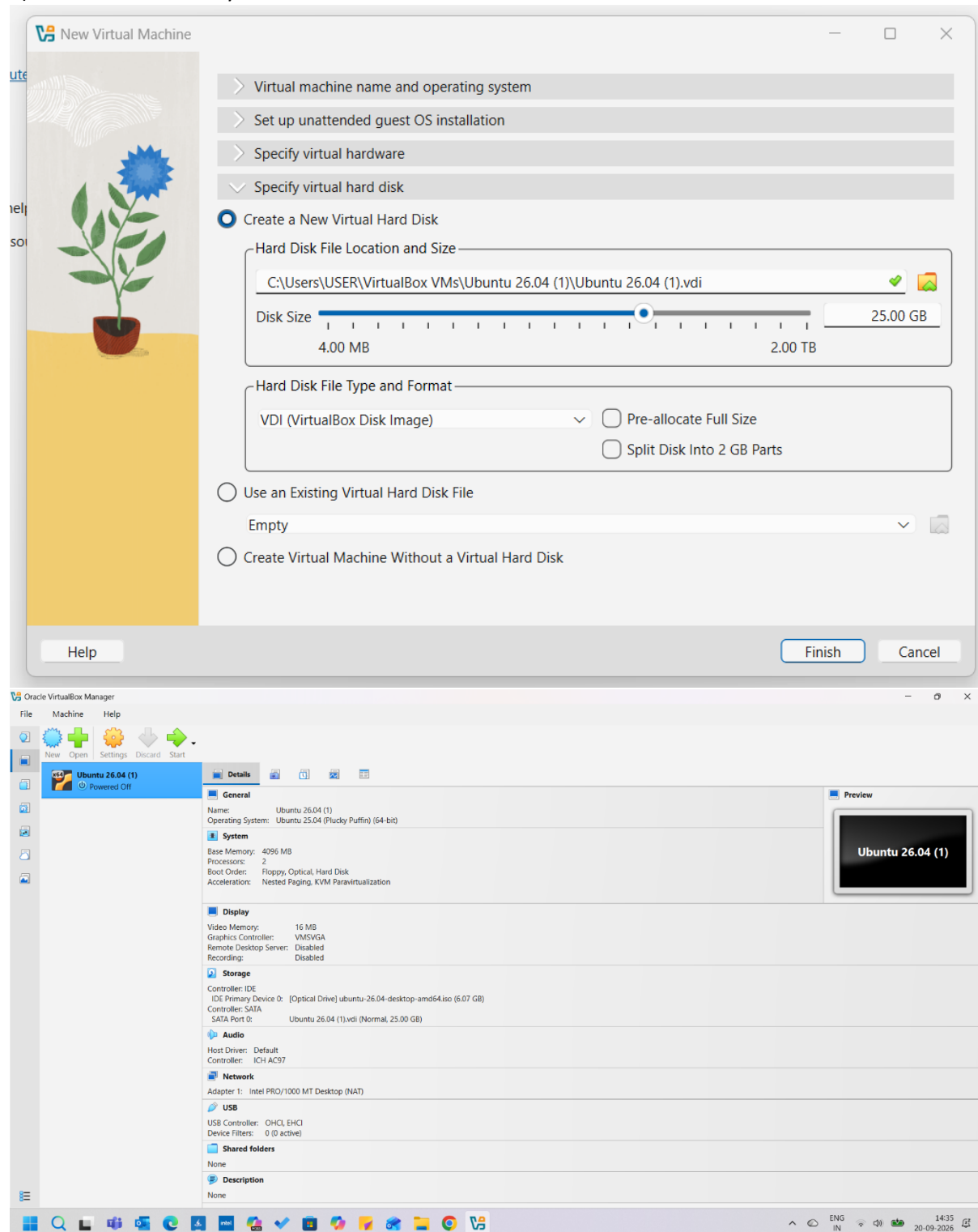
- 4) Don't do anything just let it by default



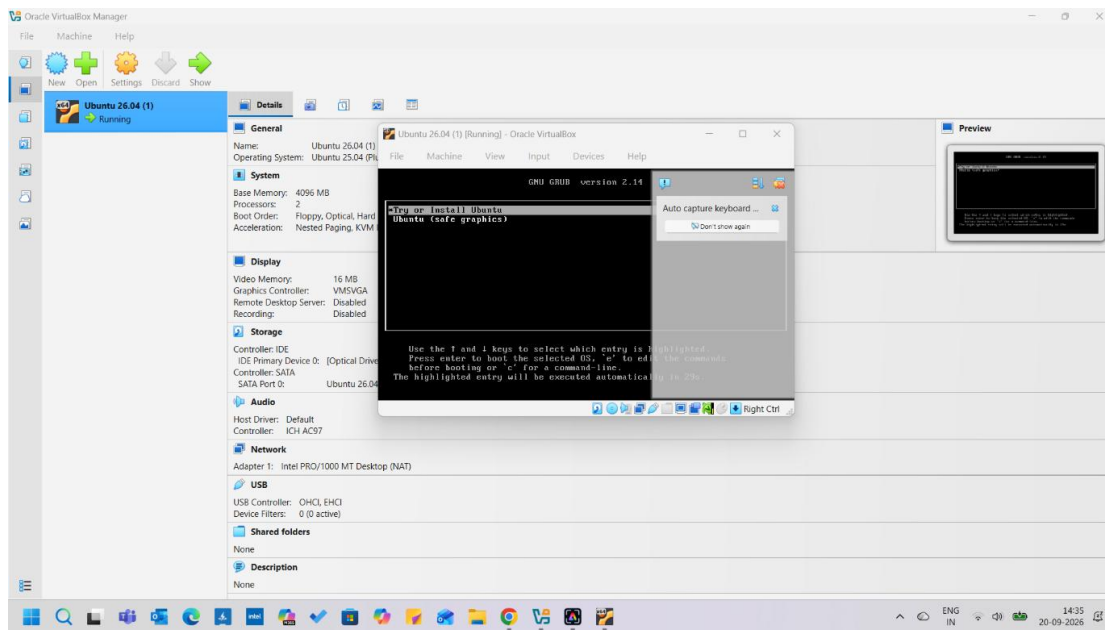
- 5) Set the memory = 4 GB or 4096 MB
- 6) Set the number of CPUs = 2



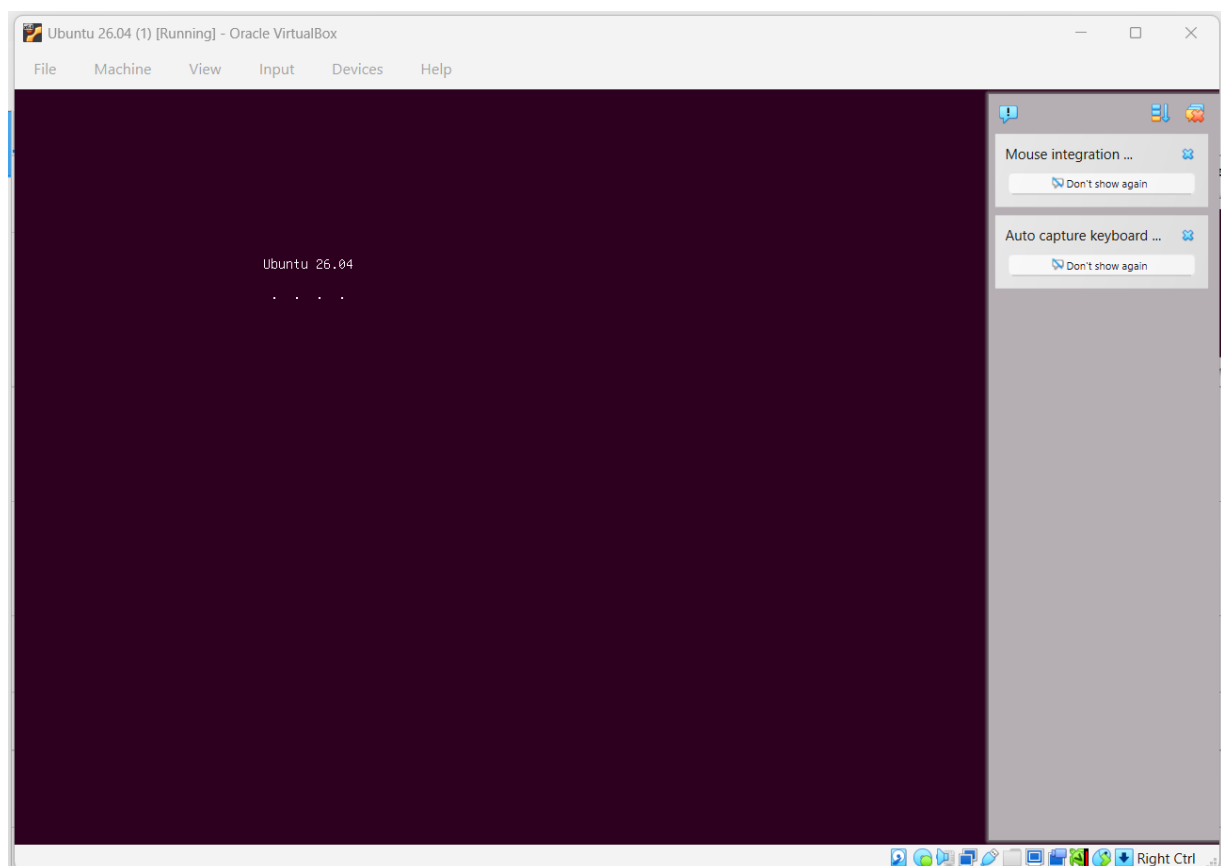
7) Allocate the memory or disc size = 25.00 GB

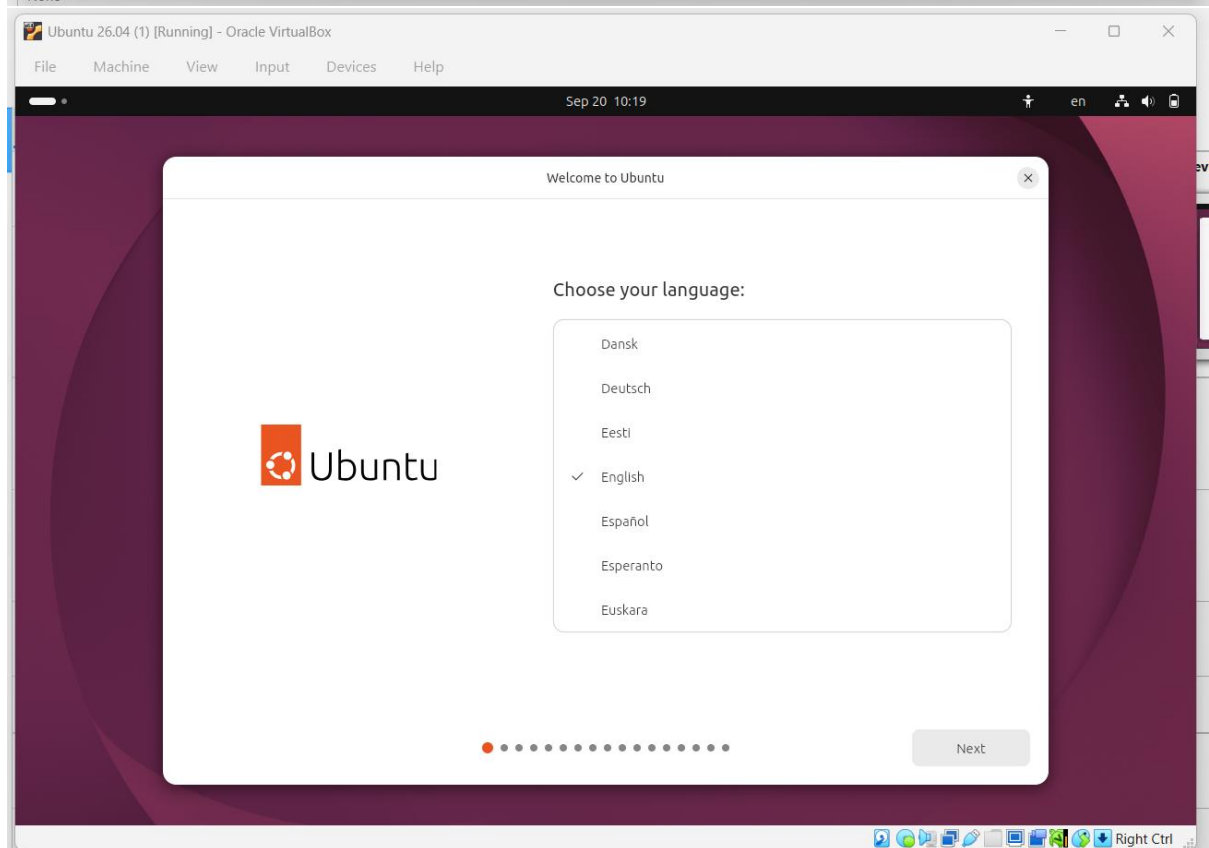
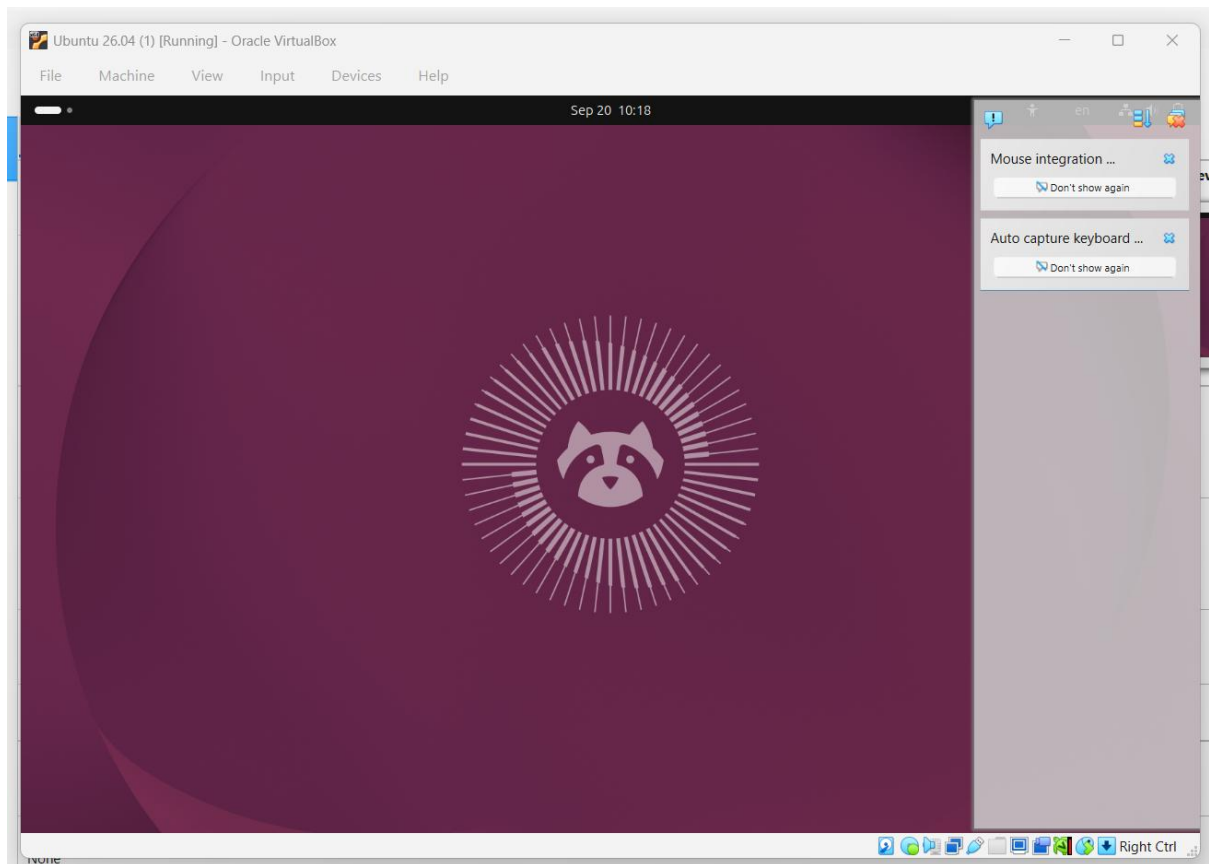


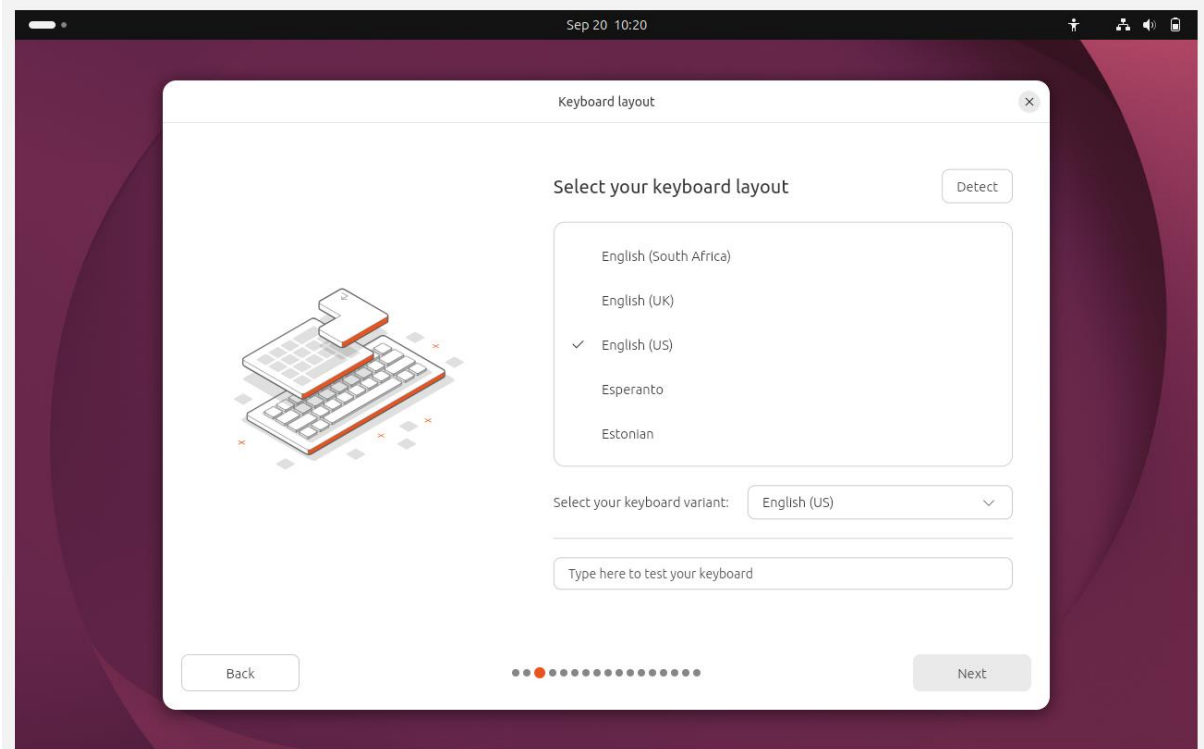
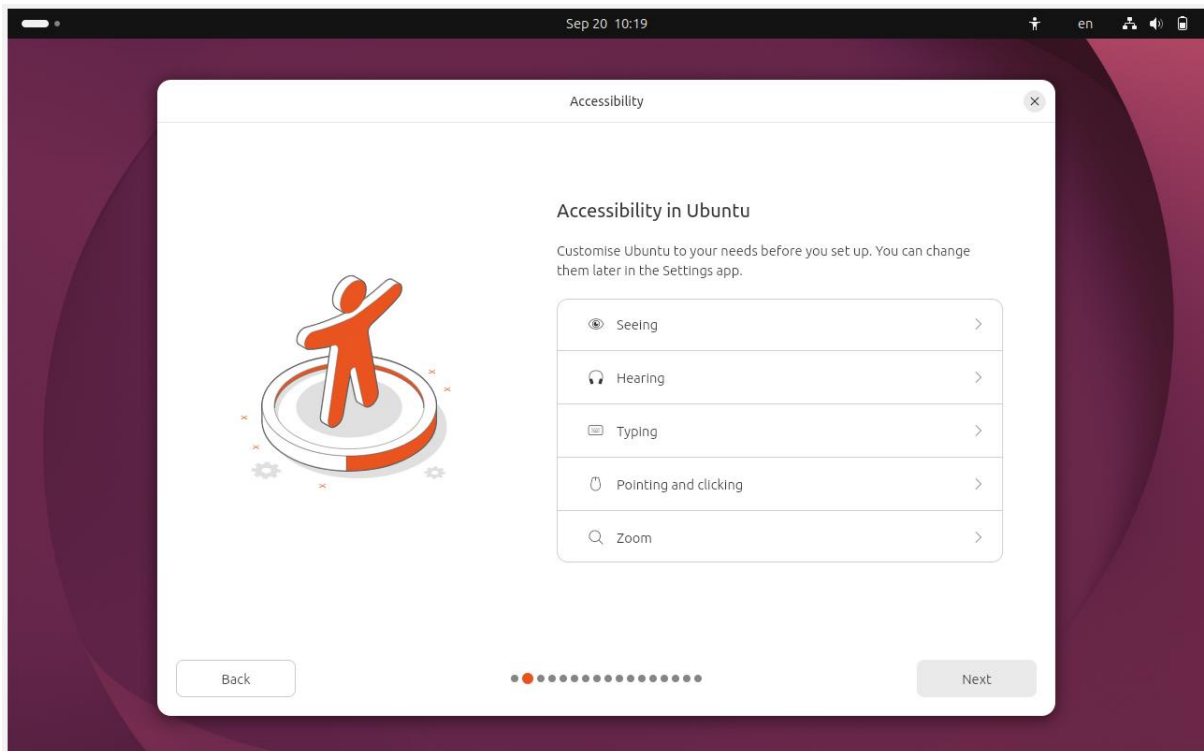
8) Now Do click on “Try or instal Ubuntu”

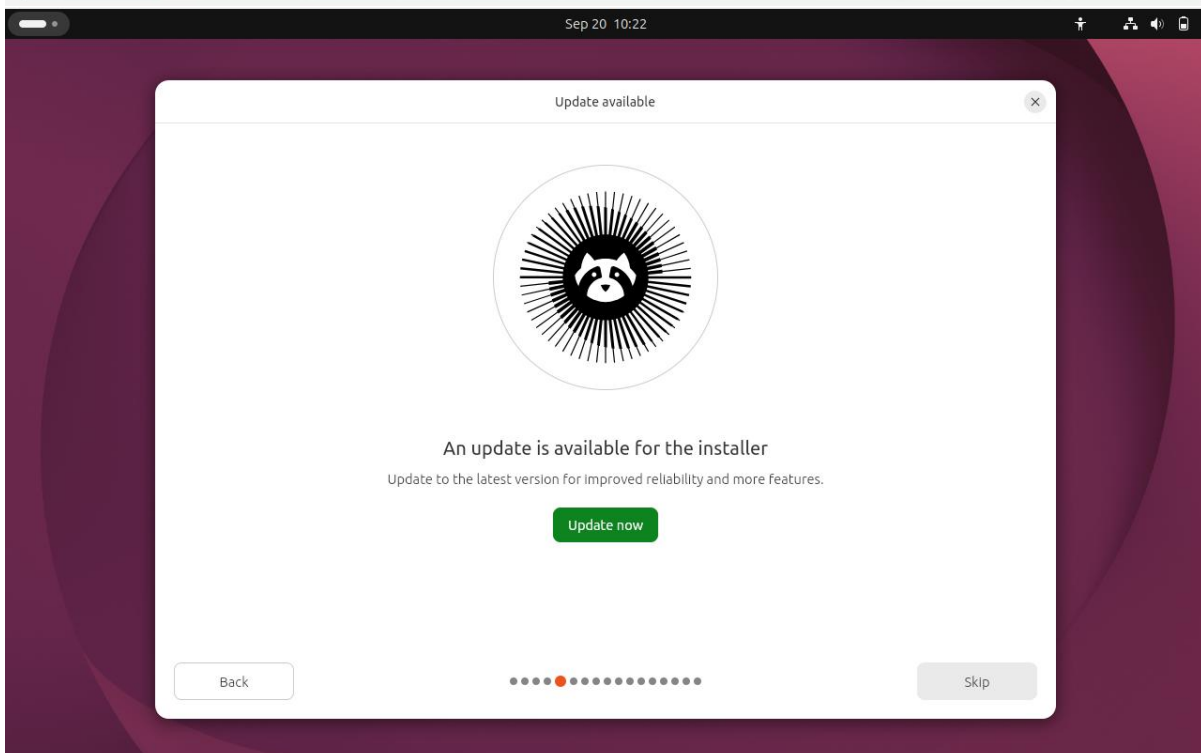
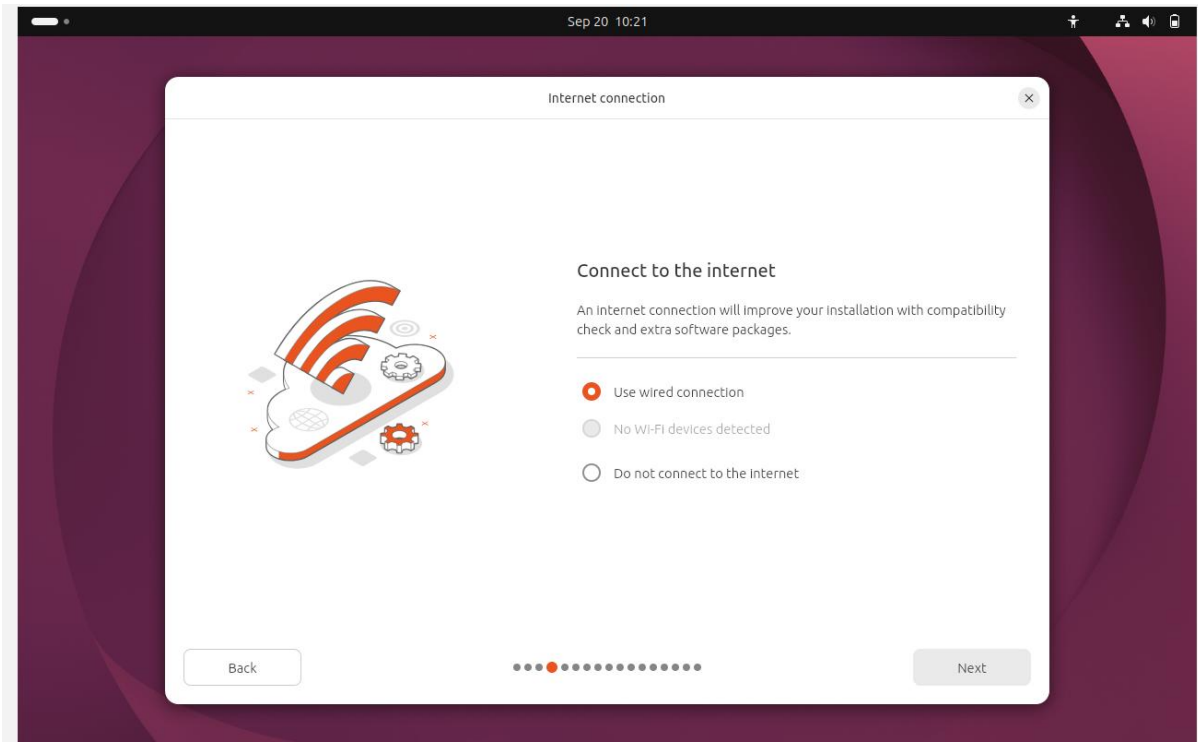


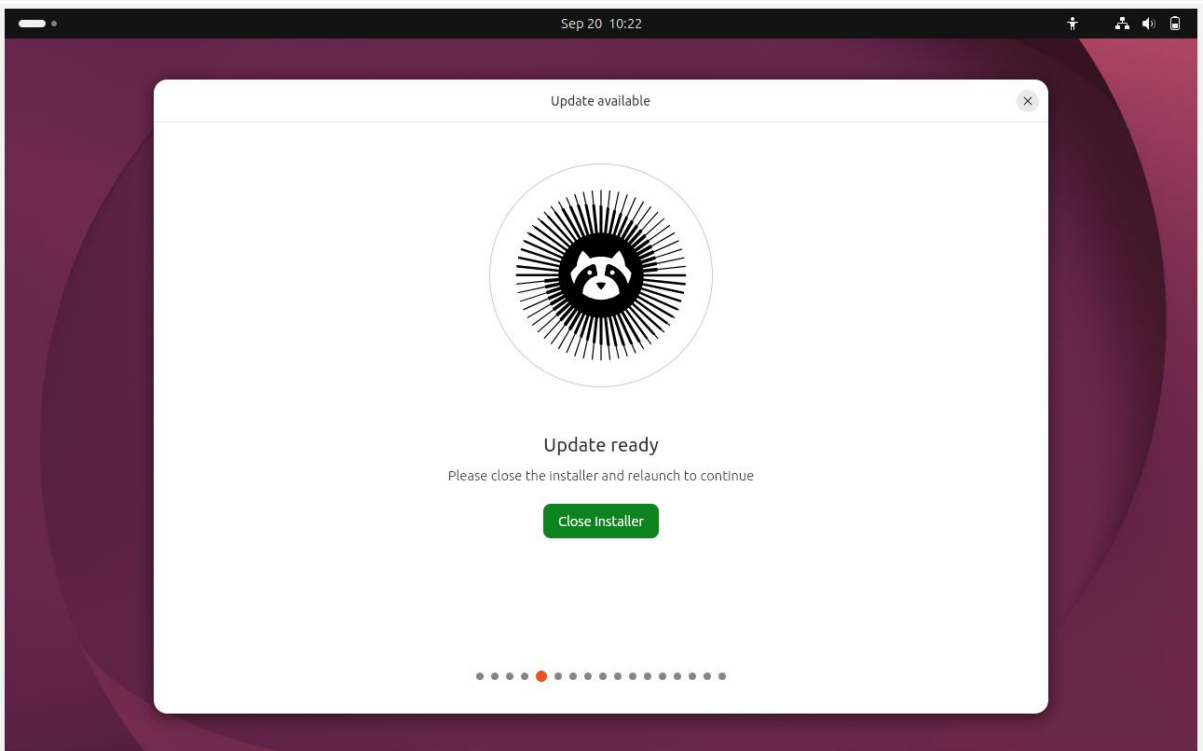
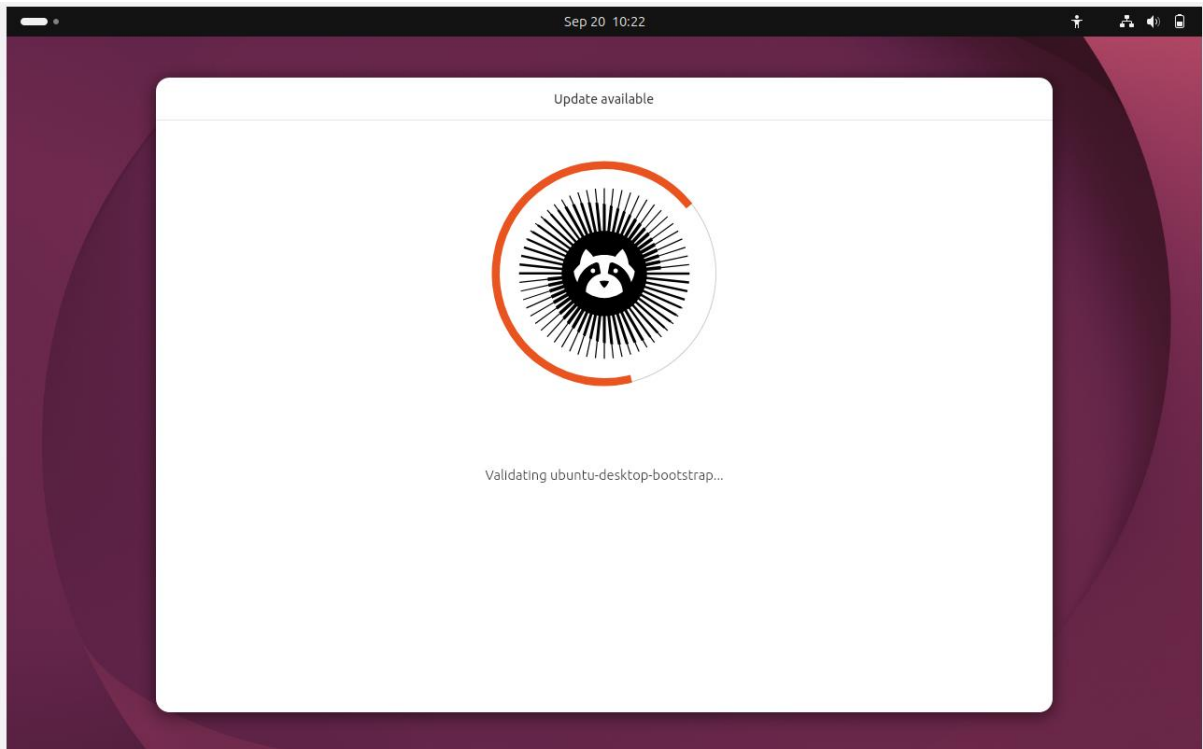
9) Now Ubuntu is Installing

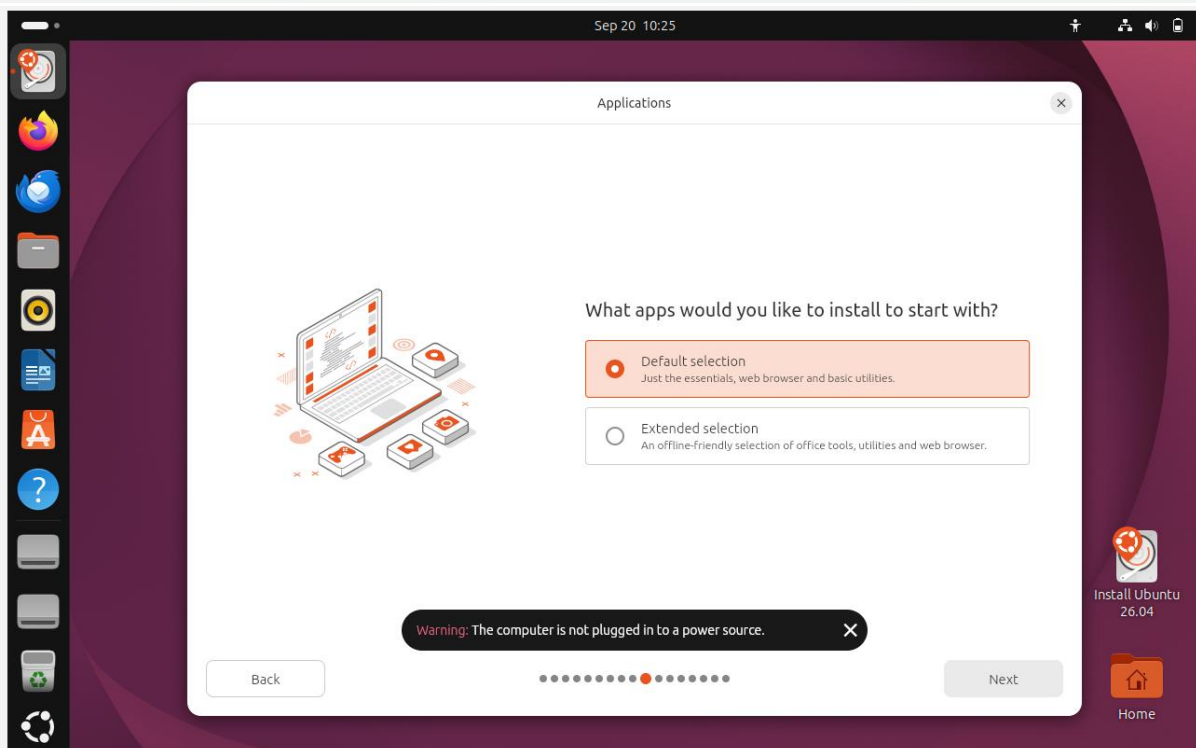
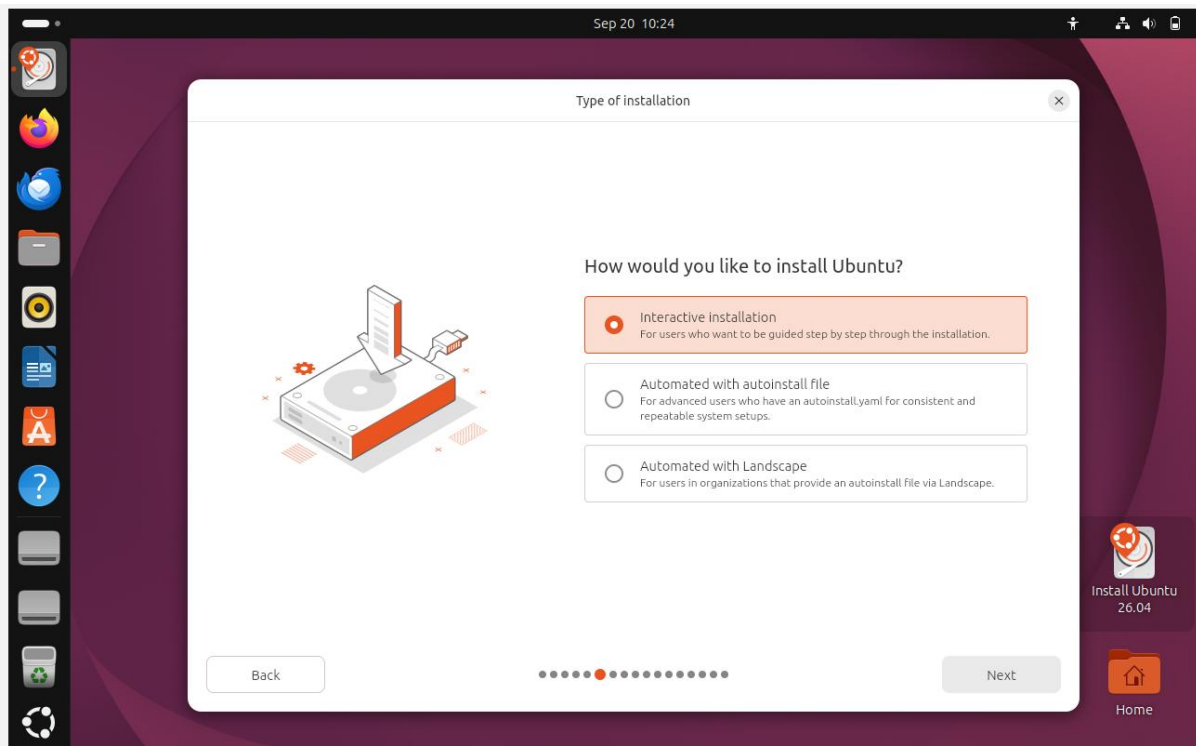


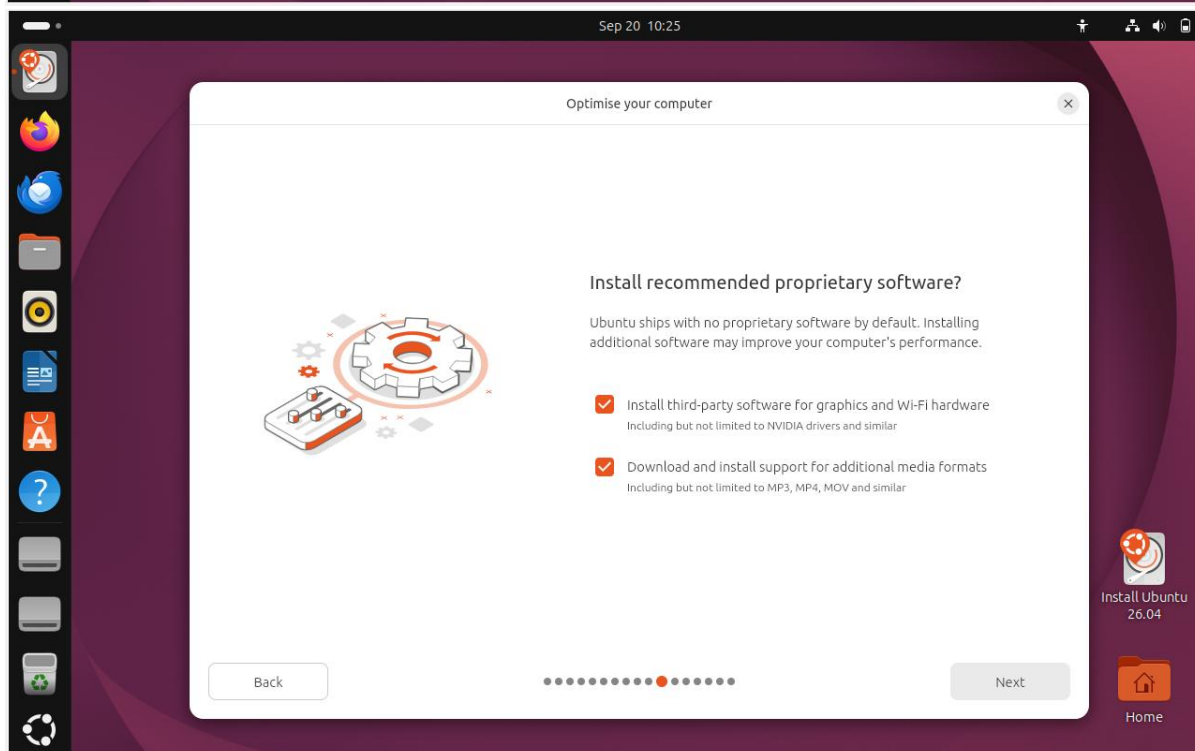
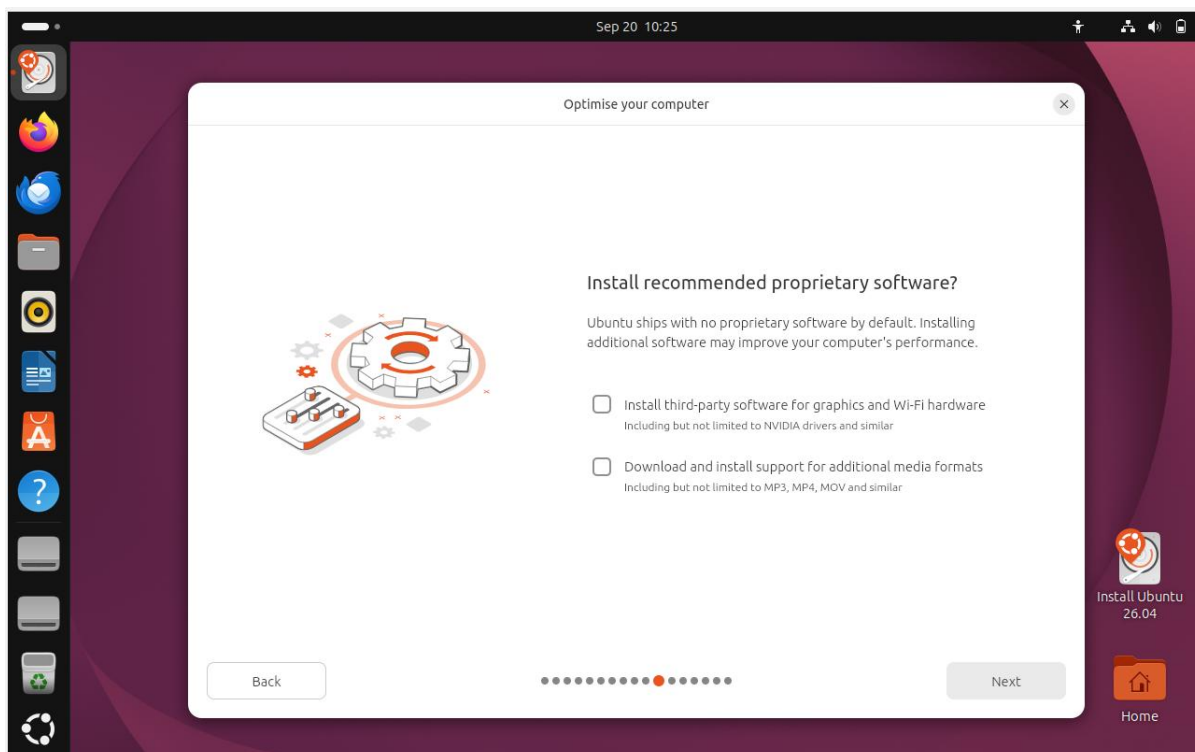


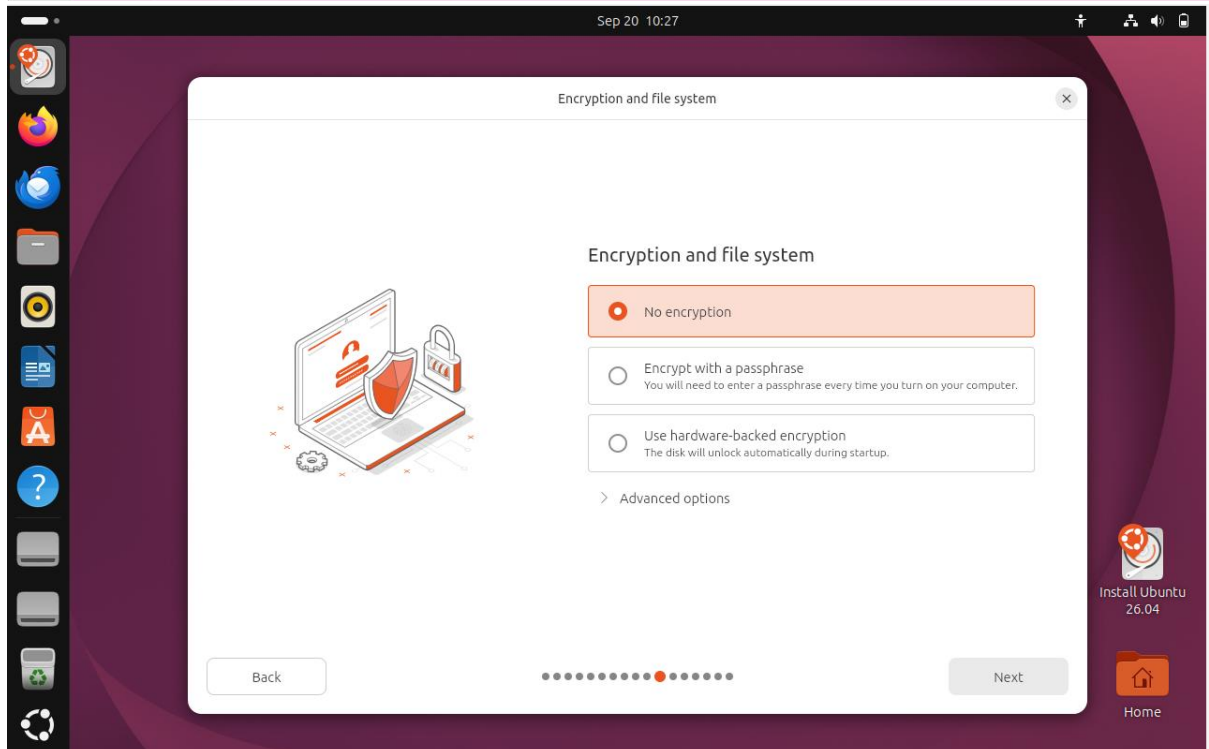
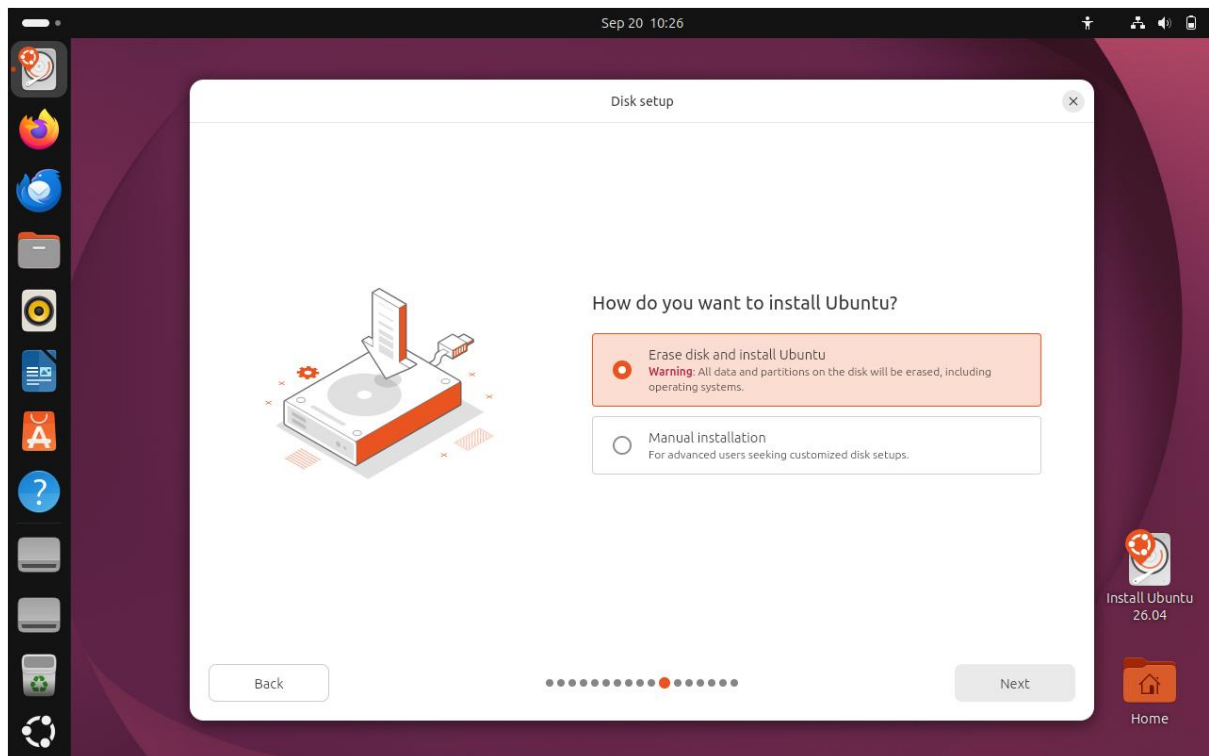


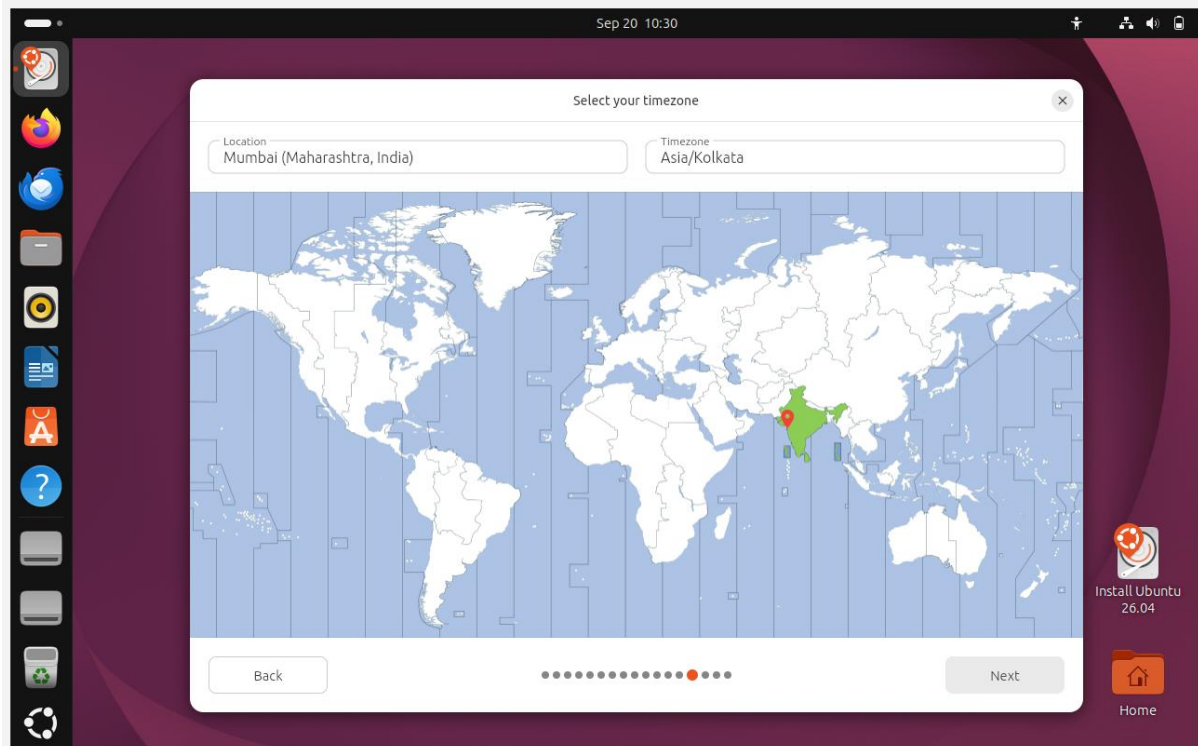
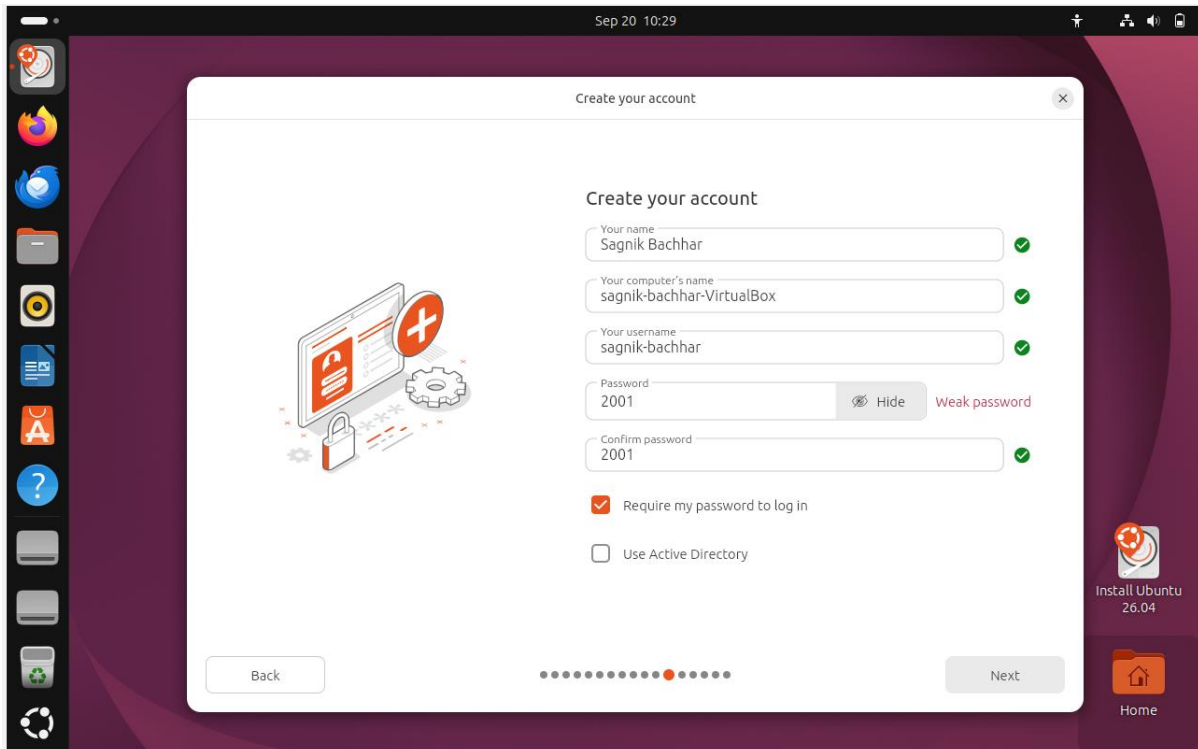


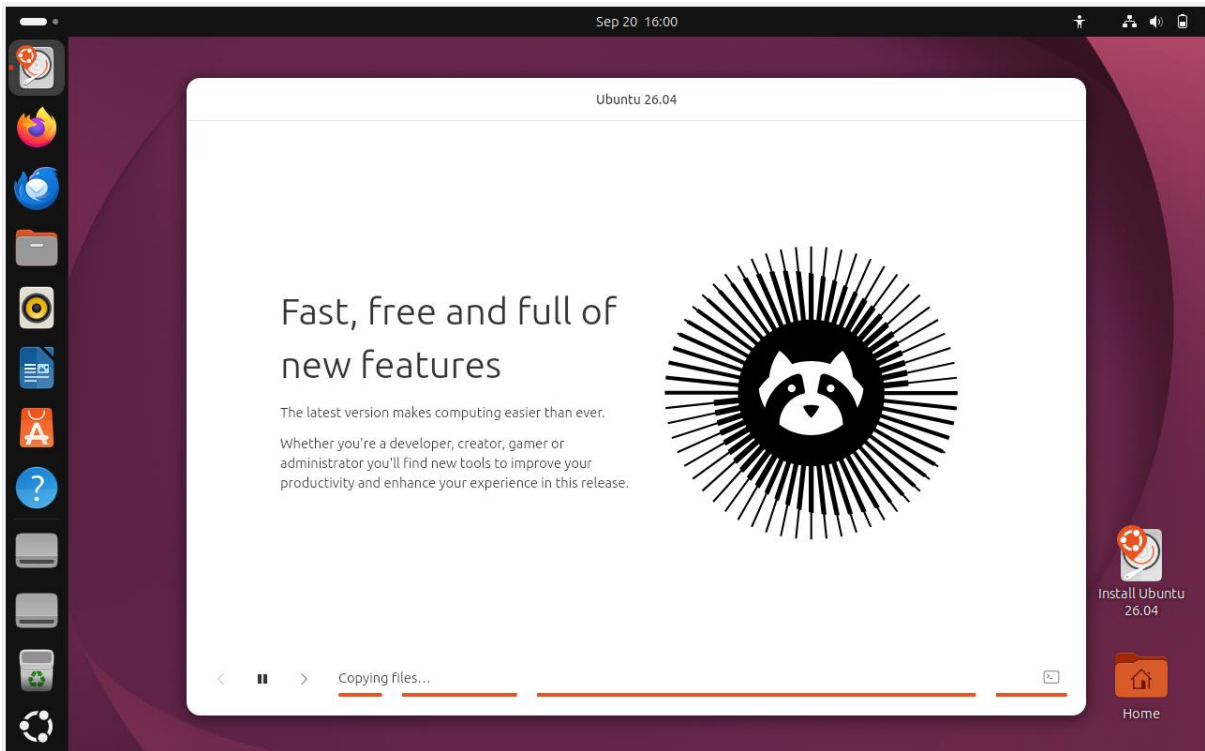
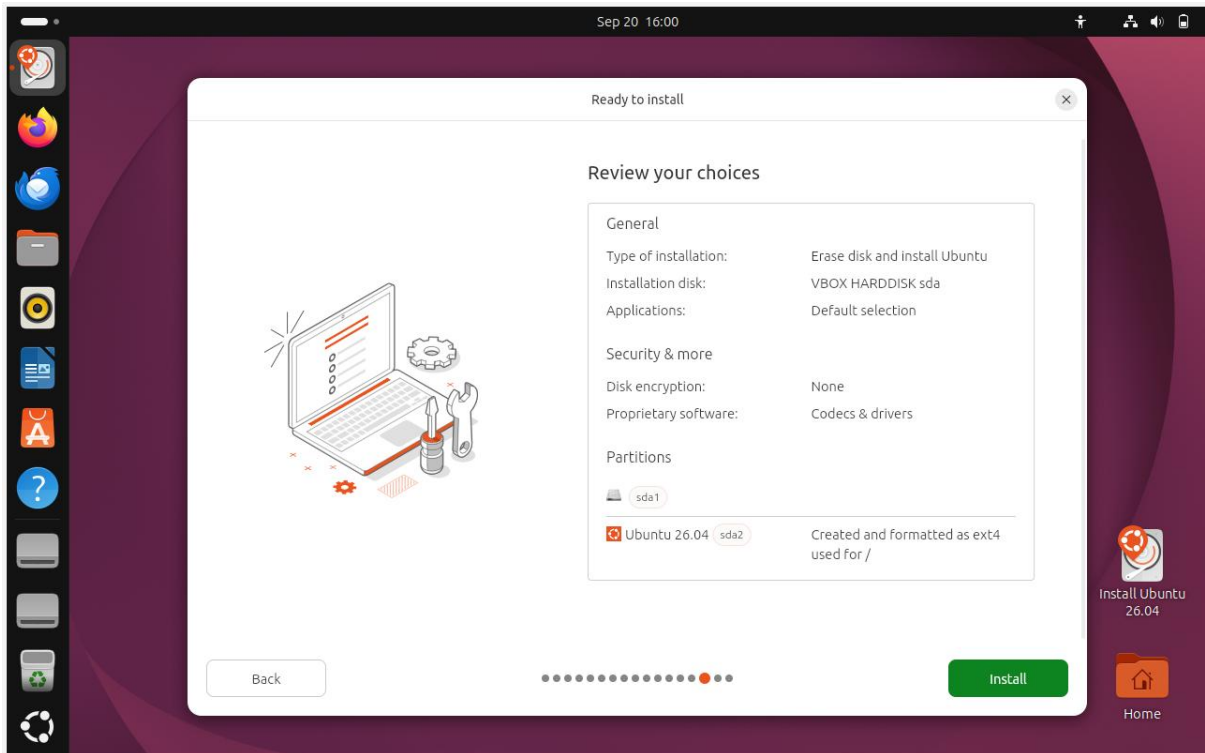


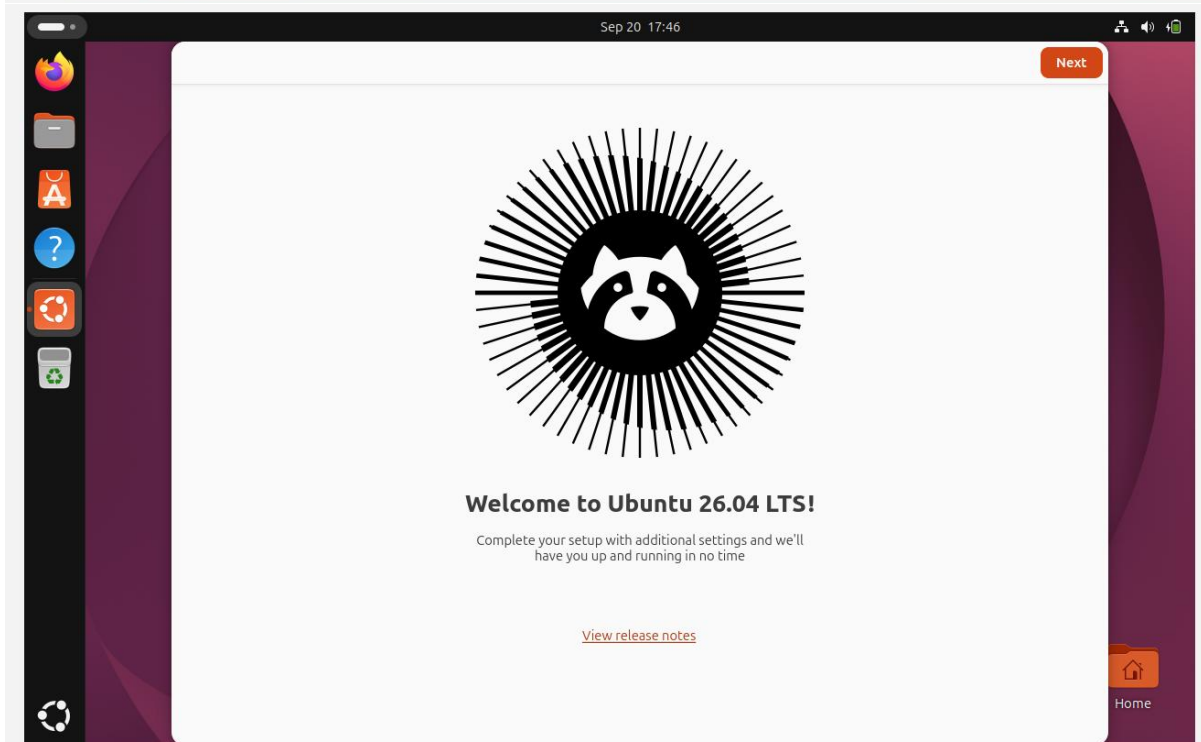
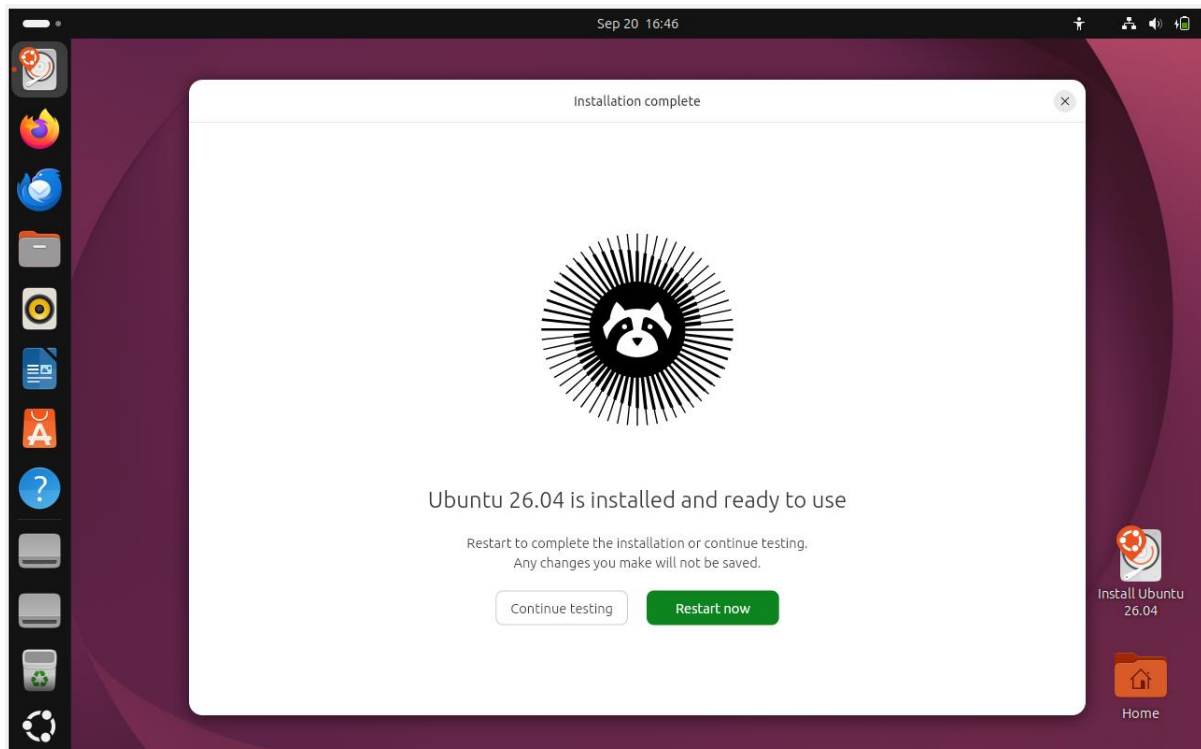


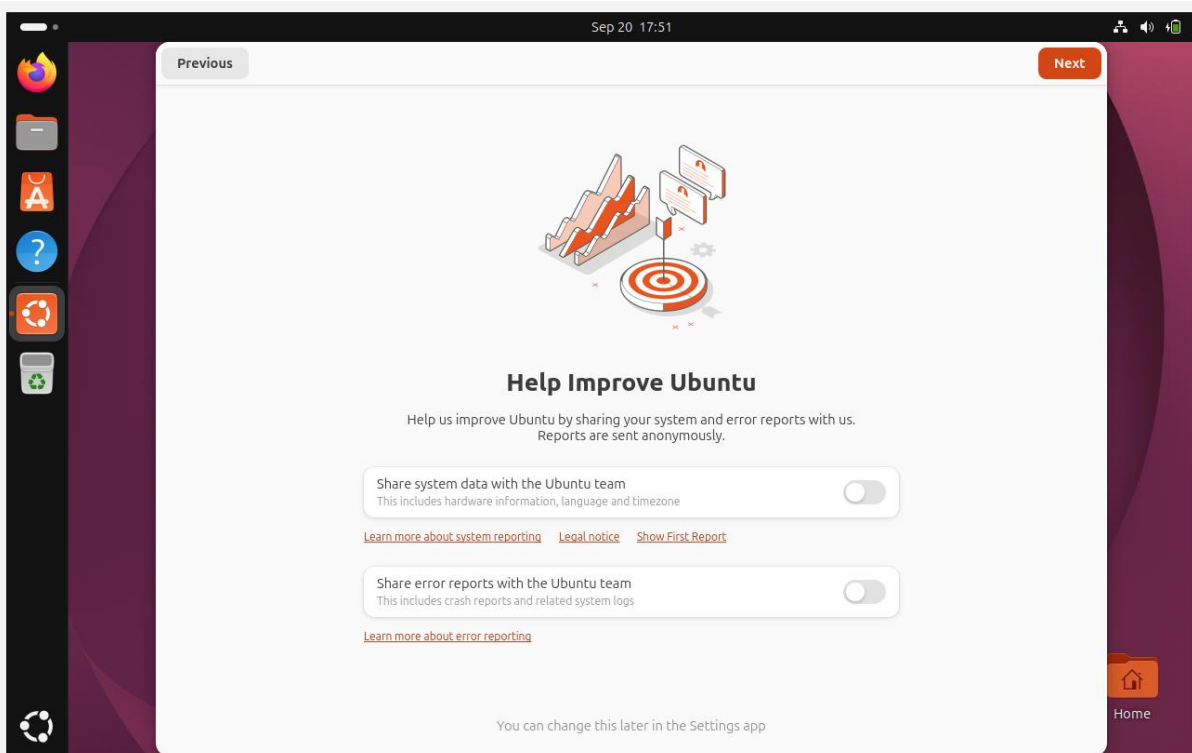
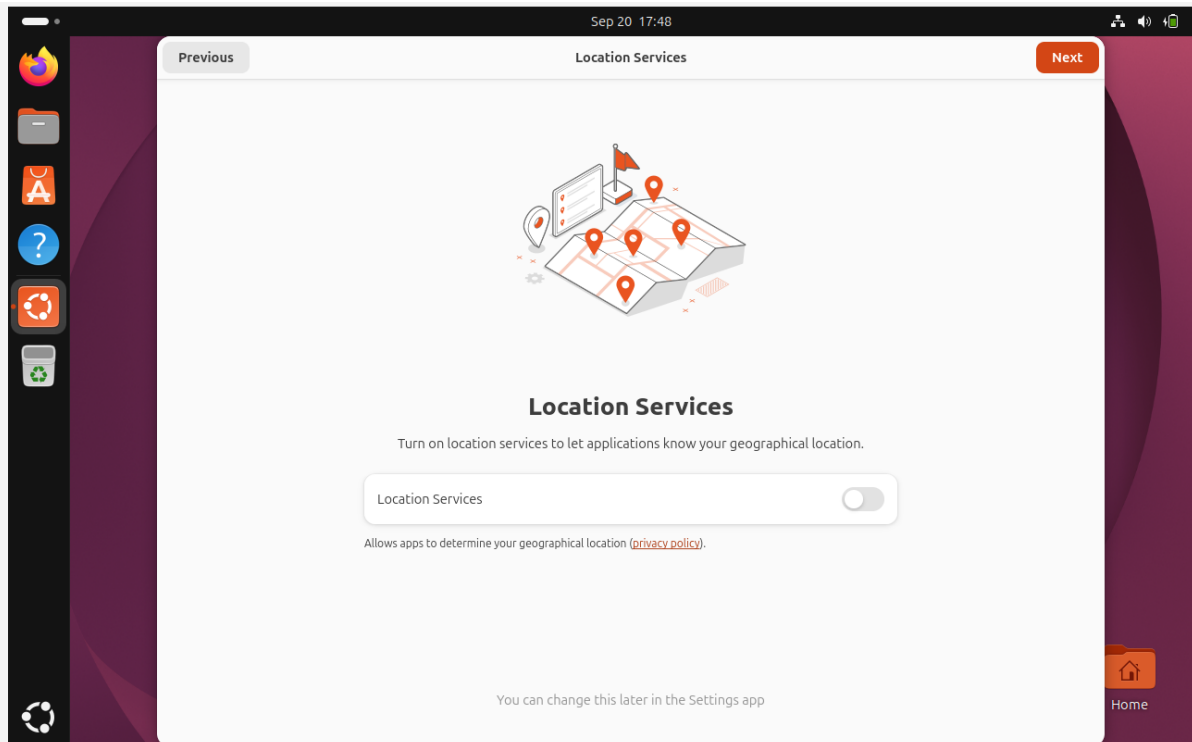


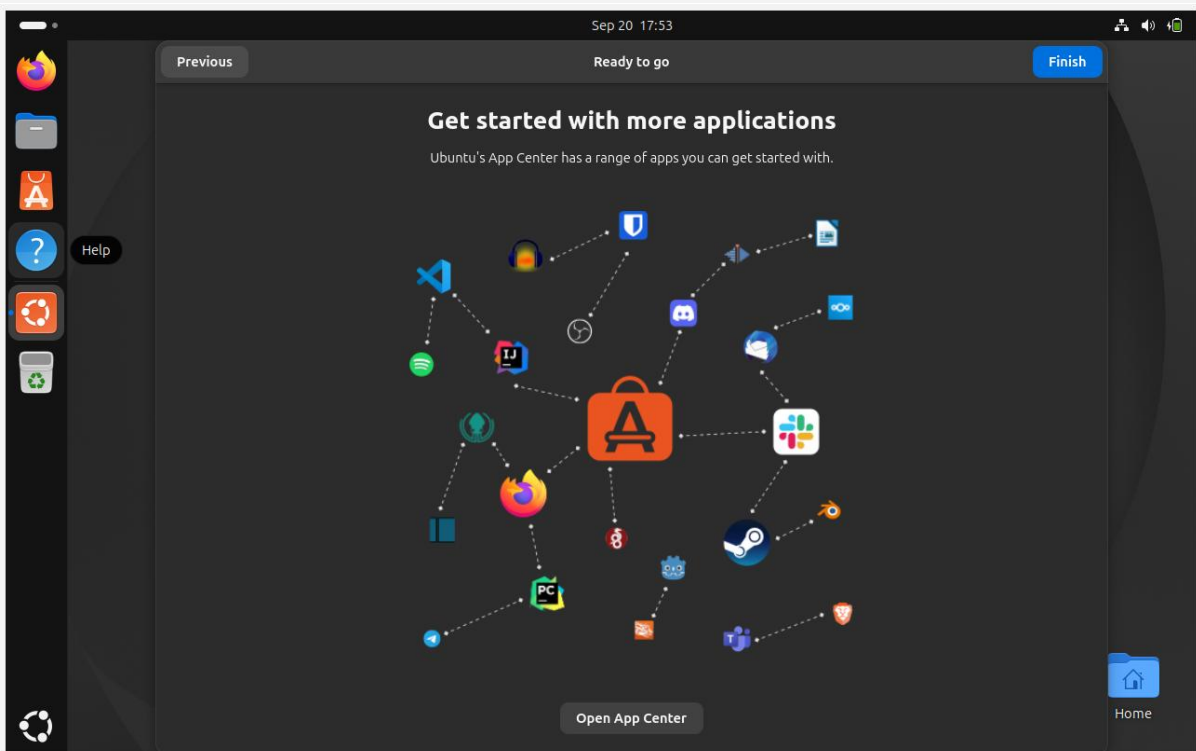
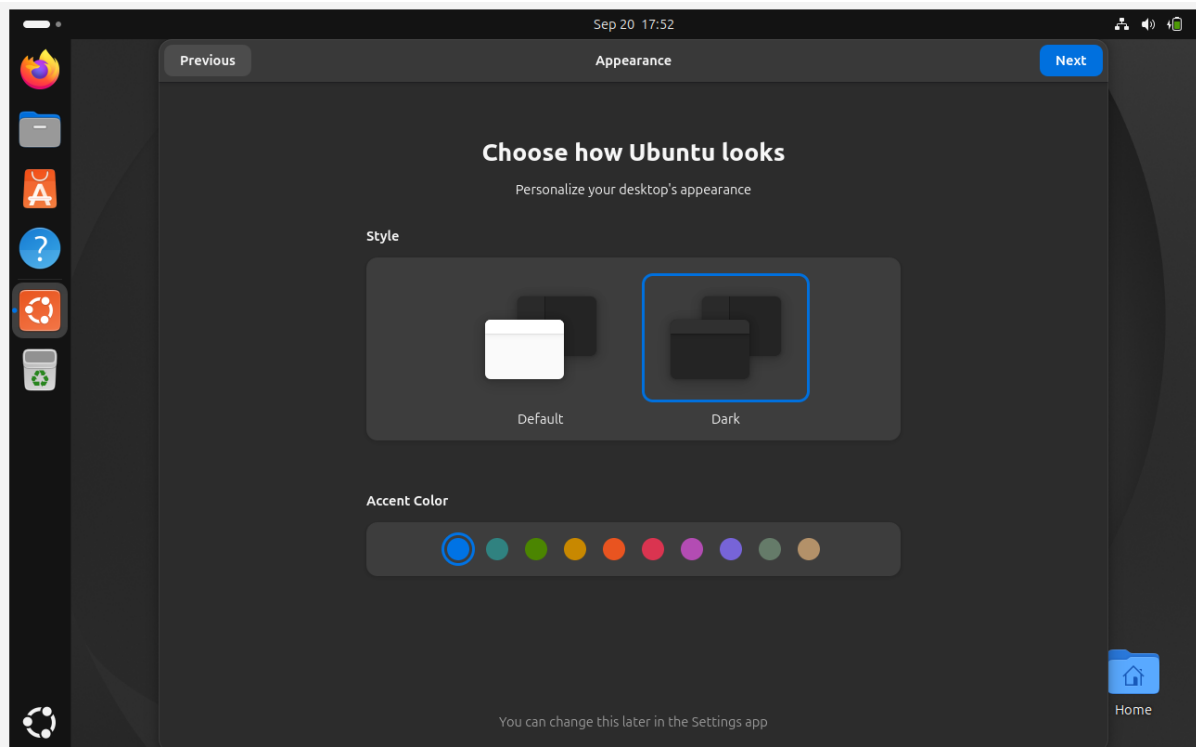


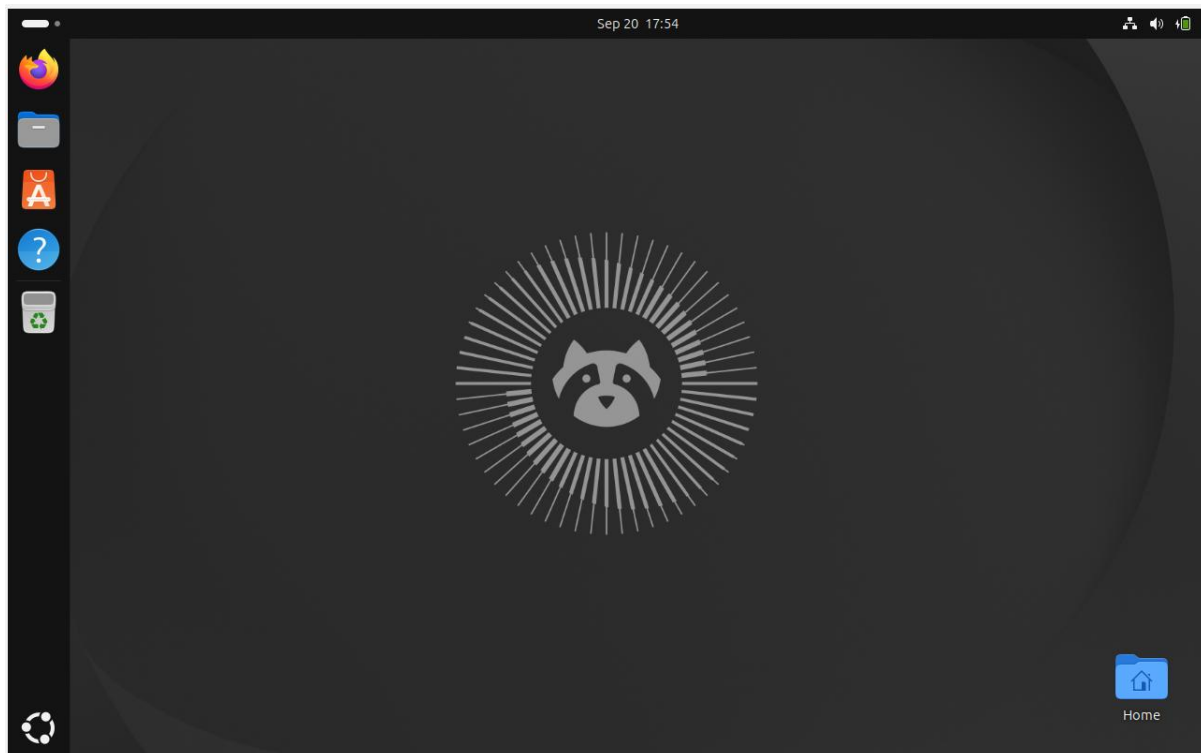






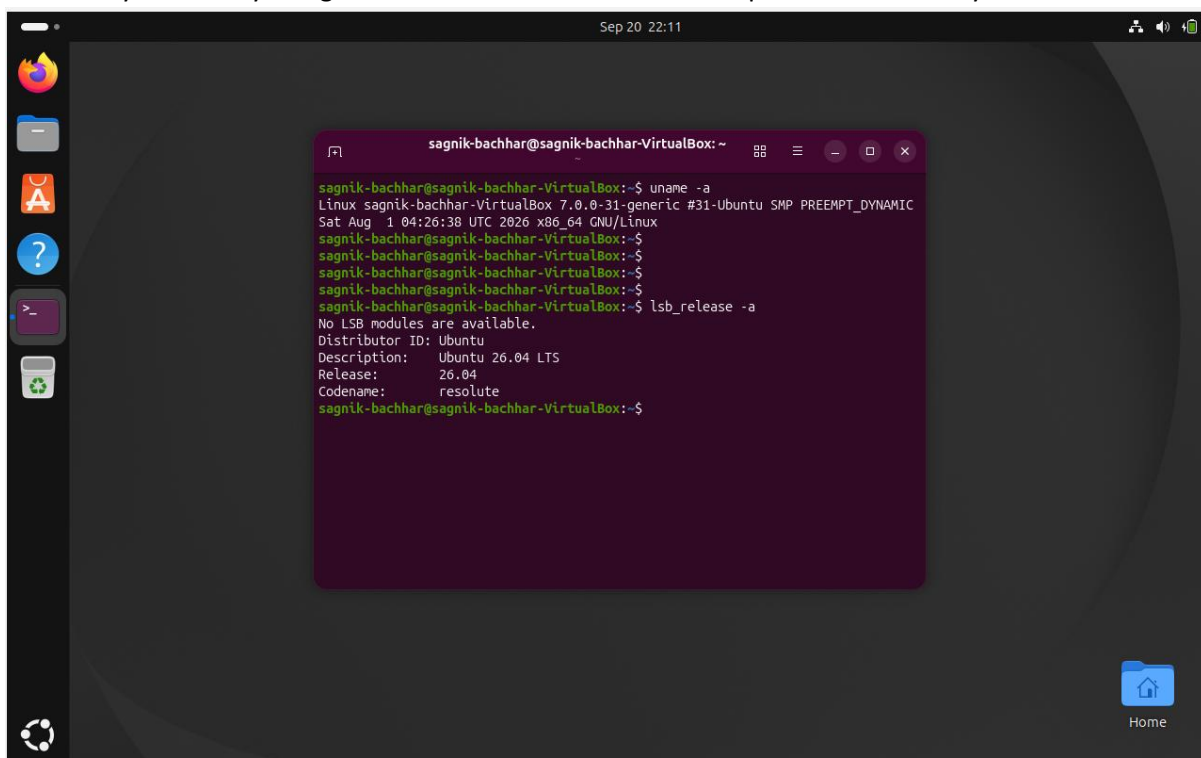






Installation of VM1 is done.

Now verify the VM by using various commands & Install the required build tools By Linux command :



```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:52:3c:55 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 82472sec preferred_lft 82472sec
    inet6 fd17:625c:f037:2:a00:27ff:fe52:3c55/64 scope global dynamic noprefixroute
        valid_lft 86372sec preferred_lft 14372sec
    inet6 fe80::a00:27ff:fe52:3c55/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip -a
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | fou | help | ila | ioam | l2tp | link |
                  macsec | maddress | monitor | mptcp | mroute | mrule |
                  neighbor | neighbour | netconf | netns | nexthop | ntable |
                  ntbl | route | rule | sr | stats | tap | tcpmetrics |
                  token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                   -h[uman-readable] | -iec | -j[son] | -p[retty] |
                   -f[amily] { inet | inet6 | mpls | bridge | link } |
                   -4 | -6 | -M | -B | -0 |
                   -l[oops] { maximum-addr-flush-attempts } | -echo | -br[ief] |
                   -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
                   -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
                   -c[olor]}
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

```

enp0s3: command not found
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:52:3c:55 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 82329sec preferred_lft 82329sec
    inet6 fd17:625c:f037:2:a00:27ff:fe52:3c55/64 scope global dynamic noprefixroute
        valid_lft 86229sec preferred_lft 14229sec
    inet6 fe80::a00:27ff:fe52:3c55/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ enp0s3
enp0s3: command not found
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$

```

```

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ dpkg -1 | grep linux-headers
dpkg: error: unknown option -1

```

Type dpkg --help for help about installing and deinstalling packages [*];
 Use 'apt' or 'aptitude' for user-friendly package management;
 Type dpkg -Dhelp for a list of dpkg debug flag values;
 Type dpkg --force-help for a list of forcing options;
 Type dpkg-deb --help for help about manipulating *.deb files;

Options marked [*] produce a lot of output - pipe it through 'less' or 'more' !

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

```

Type dpkg --help for help about installing and deinstalling packages [*];
Use 'apt' or 'aptitude' for user-friendly package management;
Type dpkg -Dhelp for a list of dpkg debug flag values;
Type dpkg --force-help for a list of forcing options;
Type dpkg-deb --help for help about manipulating *.deb files;

Options marked [*] produce a lot of output - pipe it through 'less' or 'more' !
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ dpkg -l | grep linux-headers
ii  linux-headers-7.0.0-14          7.0.0-14.14          all          Header files related to Linux kernel version 7.
0.0
ii  linux-headers-7.0.0-14-generic 7.0.0-14.14          amd64        Linux kernel headers for version 7.0.0
ii  linux-headers-7.0.0-31          7.0.0-31.31          all          Header files related to Linux kernel version 7.
0.0
ii  linux-headers-7.0.0-31-generic 7.0.0-31.31          amd64        Linux kernel headers for version 7.0.0
ii  linux-headers-generic-hwe-26.04 7.0.0-31.31          amd64        Generic Linux kernel headers
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$

```

```
Sep 21 00:10
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ uname -a
Linux sagnik-bachhar-VirtualBox 7.0.0-31-generic #31-Ubuntu SMP PREEMPT_DYNAMIC Sat Aug 1 04:26:38 UTC 2026 x86_64 GNU/Linux
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 26.04 LTS
Release: 26.04
Codename: resolute
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: GenuineIntel
Model name: Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz
CPU family: 6
Model: 142
Thread(s) per core: 1
Core(s) per socket: 2
Socket(s): 1
Stepping: 12
BogoMIPS: 4224.01
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm
constant tsc rep nopl xtopology nonstop tsc rcpuid tsc known freq npi nplmulldo ssse3 fma cx16 prid sse4_1 sse4_2 mo
```

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ uname -a
Linux sagnik-bachhar-VirtualBox 7.0.0-31-generic #31-Ubuntu SMP PREEMPT_DYNAMIC Sat Aug 1 04:26:38 UTC 2026 x86_64 GNU/Linux
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 26.04 LTS
Release: 26.04
Codename: resolute
```

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
```

```
Sep 21 00:11
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 2
  On-line CPU(s) list:  0,1
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz
CPU family:             6
Model:                 142
Thread(s) per core:     1
Core(s) per socket:     2
Socket(s):              1
Stepping:               12
BogoMIPS:               4224.01
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm
                        constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 mo
                        vbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ibrs_enhanced fsgsbase bmi1 avx2 bmi2 invpcid r
                        dseed adx clflushopt arat md_clear flush_lid arch_capabilities

Virtualization features:
Hypervisor vendor:     KVM
Virtualization type:   full
Caches (sum of all):
  L1d:                  64 KiB (2 instances)
  L1i:                  64 KiB (2 instances)
  L2:                   512 KiB (2 instances)
  L3:                   12 MiB (2 instances)
NUMA:
  NUMA node(s):         1
  NUMA node0 CPU(s):    0,1
Vulnerabilities:
  Gather data sampling:  Not affected
  Ghostwrite:           Not affected
  Indirect target selection: Mitigation; Aligned branch/return thunks
  Itlb multihit:        Not affected
  L1tf:                 Not affected
  Mds:                 Not affected
  Meltdown:            Not affected
  Mmio stale data:      Mitigation; Clear CPU buffers; SMT Host state unknown
```

```
Sep 21 00:12
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
Flags:                  tpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm
                        constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 mo
                        vbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ibrs_enhanced fsgsbase bmi1 avx2 bmi2 invpcid r
                        dseed adx clflushopt arat md_clear flush_lid arch_capabilities

Virtualization features:
Hypervisor vendor:     KVM
Virtualization type:   full
Caches (sum of all):
  L1d:                  64 KiB (2 instances)
  L1i:                  64 KiB (2 instances)
  L2:                   512 KiB (2 instances)
  L3:                   12 MiB (2 instances)
NUMA:
  NUMA node(s):         1
  NUMA node0 CPU(s):    0,1
Vulnerabilities:
  Gather data sampling:  Not affected
  Ghostwrite:           Not affected
  Indirect target selection: Mitigation; Aligned branch/return thunks
  Itlb multihit:        Not affected
  L1tf:                 Not affected
  Mds:                 Not affected
  Meltdown:            Not affected
  Mmio stale data:      Mitigation; Clear CPU buffers; SMT Host state unknown
  Old microcode:        Not affected
  Reg file data sampling: Not affected
  Retbleed:             Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass:    Vulnerable
  Spectre v1:           Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:           Mitigation; Enhanced / Automatic IBRS; PBRSE-eIBRS SW sequence; BHI SW loop, KVM SW loop
  Srbds:                Unknown: Dependent on hypervisor status
  Tsa:                  Not affected
  Tsx async abort:      Not affected
  Vmscape:              Not affected

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ lscpu | grep -E "CPU(s)|Model name"
CPU(s):                2
On-line CPU(s) list:   0,1
Model name:            Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz
NUMA node0 CPU(s):     0,1
```

```
Sep 21 00:12
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.3Gi        1.2Gi        383Mi        32Mi        2.1Gi        2.2Gi
Swap:          3.8Gi        1.4Mi        3.8Gi

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 4K 1 loop /snap/bare/5
loop1 7:1 0 273.7M 1 loop /snap/firefox/8107
loop2 7:2 0 16.5M 1 loop /snap/firmware-updater/226
loop3 7:3 0 19.6M 1 loop /snap/desktop-security-center/150
loop4 7:4 0 66.8M 1 loop /snap/core24/1587
loop5 7:5 0 395M 1 loop /snap/ mesa-2404/1165
loop6 7:6 0 91.7M 1 loop /snap/gtk-common-themes/1535
loop7 7:7 0 606.1M 1 loop /snap/gnome-46-2404/153
loop8 7:8 0 15.7M 1 loop /snap/snap-store/1367
loop9 7:9 0 18.8M 1 loop /snap/prompting-client/204
loop10 7:10 0 49.3M 1 loop /snap/snapd/26865
loop11 7:11 0 580K 1 loop /snap/snapd-desktop-integration/361
sda 8:0 0 25G 0 disk
├─sda1 8:1 0 1M 0 part
└─sda2 8:2 0 25G 0 part /
sr0 11:0 1 1024M 1 rom

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           680M  3.2M  677M   1% /run
/dev/sda2       25G   11G   13G  47% /
tmpfs           1.7G   0 1.7G   0% /dev/shm
tmpfs           1.7G  24K  1.7G   1% /tmp
tmpfs           340M   88K  340M   1% /run/user/1000
none            1.0M   0 1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:52:3c:55 brd ff:ff:ff:ff:ff:ff
```

```
Sep 21 00:13
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:52:3c:55 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 77060sec preferred_lft 77060sec
    inet6 fd17:625c:f037:2:a00:27ff:fe52:3c55/64 scope global dynamic noprefixroute
        valid_lft 86028sec preferred_lft 14028sec
    inet6 fe80::a00:27ff:fe52:3c55/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ping -c 4 google.com
PING google.com (142.251.221.110) 56(84) bytes of data:
64 bytes from lcnaaa-az-in-f14.1e100.net (142.251.221.110): icmp_seq=1 ttl=64 time=49.1 ms
64 bytes from lcnaaa-az-in-f14.1e100.net (142.251.221.110): icmp_seq=2 ttl=64 time=50.3 ms
64 bytes from lcnaaa-az-in-f14.1e100.net (142.251.221.110): icmp_seq=3 ttl=64 time=51.1 ms
64 bytes from lcnaaa-az-in-f14.1e100.net (142.251.221.110): icmp_seq=4 ttl=64 time=52.1 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3001ms
rtt min/avg/max/mdev = 49.119/50.655/52.066/1.076 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ make --version
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ls /usr/src/linux-headers-$(uname -r)
Documentation  Kconfig  Module.symvers  block  crypto  fs  init  ipc  lib  net  samples  security  tools  usr
```

```
Sep 21 00:13
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ make --version
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ls /usr/src/linux-headers-$(uname -r)
Documentation  Kconfig  Module.symvers  block  crypto  fs  init  ipc  lib  net  samples  security  tools  usr
Kbuild  Makefile  arch  certs  drivers  include  io_uring  kernel  mm  rust  scripts  sound  ubuntu  virt
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ dpkg -l | grep linux-headers
ii  linux-headers-7.0.0-14  7.0.0-14.14  all  Header files related to Linux kernel version 7.0.0
ii  linux-headers-7.0.0-14-generic  7.0.0-14.14  amd64  Linux kernel headers for version 7.0.0
ii  linux-headers-7.0.0-31  7.0.0-31.31  all  Header files related to Linux kernel version 7.0.0
ii  linux-headers-7.0.0-31-generic  7.0.0-31.31  amd64  Linux kernel headers for version 7.0.0
ii  linux-headers-generic-hwe-26.04  7.0.0-31.31  amd64  Generic Linux kernel headers
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== OS ===="
==== OS ====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ lsb_release -d
Description:  Ubuntu 26.04 LTS
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Kernel ===="
==== Kernel ====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ uname -r
7.0.0-31-generic
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== CPU ===="
==== CPU ====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nproc
2
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Memory ===="
==== Memory ====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.3Gi       1.3Gi       253Mi        33Mi       2.1Gi       2.0Gi
Swap:          3.8Gi       1.4Mi       3.8Gi
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Stroage ===="
==== Stroage =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            680M  3.2M  677M   1% /run
/dev/sda2       25G   11G   13G  47% /
tmpfs            1.7G   0  1.7G   0% /dev/shm
tmpfs            1.7G  24K  1.7G   1% /tmp
tmpfs            340M  88K  340M   1% /run/user/1000
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
```

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== OS ===="
==== OS =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ lsb_release -d
Description:  Ubuntu 26.04 LTS
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Kernel ===="
==== Kernel =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ uname -r
7.0.0-31-generic
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== CPU ===="
==== CPU =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nproc
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sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Memory ===="
==== Memory =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.3Gi       1.3Gi       253Mi        33Mi       2.1Gi       2.0Gi
Swap:          3.8Gi       1.4Mi       3.8Gi
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Stroage ===="
==== Stroage =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            680M  3.2M  677M   1% /run
/dev/sda2       25G   11G   13G  47% /
tmpfs            1.7G   0  1.7G   0% /dev/shm
tmpfs            1.7G  24K  1.7G   1% /tmp
tmpfs            340M  88K  340M   1% /run/user/1000
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== GCC ===="
==== GCC =====
```

```
Sep 21 00:13
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

==== CPU ====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nproc
2
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Memory ====="
==== Memory =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.3Gi          1.3Gi        253Mi        33Mi         2.1Gi         2.0Gi
Swap:          3.8Gi          1.4Mi        3.8Gi

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Stroage ====="
==== Stroage =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            680M  3.2M  677M   1% /run
/dev/sda2        25G   11G   13G   47% /
tmpfs            1.7G   0     1.7G   0% /dev/shm
tmpfs            1.7G  24K   1.7G   1% /tmp
tmpfs            340M   88K  340M   1% /run/user/1000
none             1.0M   0     1.0M   0% /run/credentials/systemd-resolved.service
none             1.0M   0     1.0M   0% /run/credentials/systemd-journald.service

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== GCC ====="
==== GCC =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ gcc --version | head -1
error: unexpected argument '-l' found

    tip: to pass '-l' as a value, use '-- -l'

Usage: head [FLAG]... [FILE]...

For more information, try '--help'.
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== GCC ====="
==== GCC =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ gcc --version | head -1
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Make ====="
==== Make =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ make --version | head -1
GNU Make 4.4.1
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== GCC ====="
==== GCC =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ gcc --version | head -1
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ echo "==== Make ====="
==== Make =====
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ make --version | head -1
GNU Make 4.4.1
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```